A Review of Related Work on Machine Learning in Semiconductor Manufacturing and Assembly Lines

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ABSTRACT

This paper deals with applications of machine learning al- gorithms in manufacturing. Machine learning can be de- fined as a field of computer science that gives computers the ability to learn without explicitly developing the needed al- gorithms. Manufacturing is the production of merchandise by manual labour, machines and tools. The focus of this paper is on automatic production lines. The areas of in- terest of this paper are semiconductor manufacturing and production on assembly lines. The purpose of this paper is to review the relevant papers describing the applications of machine learning techniques in these fields of manufacturing thus creating a firm foundation for further research in the matter of machine learning in manufacturing.

CCS Concepts

⚫Computing methodologies → Machine learning; Su- pervised learning; Unsupervised learning; ⚫Applied computing → Computer-aided manufacturing;

Keywords

machine learning, manufacturing, supervised, unsupervised, assembly line, semiconductor

1. INTRODUCTION

Machine learning applications have been present in man- ufacturing for the last two decades. Systems based on this technology are deployed with the goal to automate some of the tasks emerging from the dynamic field of industrial manufacturing. Some of the examples are expert systems for decision making support, systems for scheduling of con- current production on assembly line, systems for predictive

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maintenance of machines used in production process as well as the manufacturing fault diagnoses systems.

In order to build a knowledge-based system that auto- mates some of the tasks in manufacturing, detailed domain- expert knowledge has to be gathered. This step is crucial for any knowledge-based system as this expert knowledge is required for the system implementation. This step is also the most time-consuming and there is always the danger of invalid or incomplete knowledge transfer between the do- main expert and the developers of the system. Machine learning techniques may decrease the development time of such systems and they often reveal knowledge that might be overlooked by those acquiring the domain-knowledge.

Machine learning can be defined as a field of study in computer science that enables the personal computers to automatically get more efficient at a given task through ex- perience [37]. This field emerged from *pattern recognition* and *statistical inference*. A large portion of machine learn- ing algorithms are classification algorithms. For the given training data set, defined as classified examples, the selected algorithm develops a model which is then used to classify new datasets. Many manufacturing problems belong to the class of classification problems where the industrial domain experts are requested to assign a class to an object or dataset according to the state of the parameters of that object.

Based on the experience made in this field, faults happen quite often in the process of production of any kind. Not being able to detect and correct those faults means increase of production costs and it could even be a reason for produc- tion delay or complete standstill. These reasons led to in- creased interest of industry for machine learning techniques as a most efficient way to develop an expert knowledge-based system. The investigation of published papers in this field of study showed that machine learning techniques are used in different branches of industry. Industry fields in focus of this paper are semiconductor manufacturing and automated assembly lines.

The goal of this paper is to show through several practi- cal examples that the application of machine learning tech- niques on manufacturing problems can lead to increased pro- ductivity and decrease of production costs by early detection of production faults. The most of the publications reviewed and cited in this paper are up to 15 years old, not including the ones dealing with the theoretical concepts of machine learning. The ACM and IEEE libraries were searched for publications in the fields of interest for this paper.

One practical example of a challenge in semiconductor manufacturing is the thickness prediction of CVD (Chem- ical Vapour Deposition) on wafers. Using Virtual Metrol- ogy (VM) and Root Cause Analysis (RCA) (explained in detail in

[39]) one can detect irregularities in CVD mate- rial thickness, thus predicting the wafers of lower quality. In semiconductor manufacturing, production is based on wafers where wafers are organized into lots (25 wafers 1 lot). The goodness of the production process is assessed by measur- ing one or more parameters on the wafer (for CVD, it is the thickness of the deposited material). Such measurements are costly and time-consuming and the common practice is to do measurements only on a small portion of wafers belonging to the same lot, usually only on a single wafer. Thus, the infor- mation about the goodness of the non-assessed wafers across the lot is missing. This leads to difficulties in detecting drifts in production. One way around this problem is the Virtual Metrology approach. The production data, originating from fabrication machines from different stages of semiconductor manufacturing process (temperature, pressure, etc.) is used to estimate the goodness of physically non-assessed wafers. It provides at least an estimation of wafer quality.

2. INDUSTRIES OVERVIEW

The focus of this paper is on the application of machine learning methods in semiconductor manufacturing and as- sembly lines. This section provides rather short presentation of facts important for both of the named industry fields.

2.1 Semiconductors manufacturing

The semiconductor manufacturing industry is one of the most technologically advanced industries today and as such it is also one of the most cost-intensive industries. The $336 billion industry of semiconductors [46] provides enough op- portunities for researches to apply new technologies with the goal to decrease the production costs. The pervasive nature of the semiconductor devices implies that they are widely used in every segment of our lives; such devices can be found in our mobile phones, personal computers, cars etc. With ever increasing time-to-market expectations, solving production problems and increasing yield in such a complex production process as it is in case of a semiconductor man- ufacturing, is getting more difficult. A range of production problems can be detected by means of statistical analysis [41] and design of experiments [6] which provides a solid ground for well-tuned manufacturing processes. When the occurring problems are caused by non-linear interactions of different process parameters in combination with ever in- creasing amounts of data produced in the process, solving these problems is getting tougher. In the milestone paper [13] the authors presented the future challenges for modeling and control in semiconductor manufacturing. Since the year 2000 significant research effort is invested in this area and advances made are mostly enabled by the application of ma- chine learning and computational advances [39]. As already stated in the introductory part of this section, the semicon- ductor manufacturing is one of the most complex and tech- nologically advanced manufacturing processes. The process typically consists of more than 500 steps. All those steps in semiconductor fabrication in fab are monitored thus gener- ating immense amounts of data. In recent years, all the fab- rication equipment is delivered with equipment/production sensors. Although the real-time monitoring of

production is possible, the amount of generated data is so overwhelm- ing that the timely detection of production faults is difficult

to achieve. It is the *BigData* technology that enables the predictive maintenance in this case. The typical order of fabrication steps, as described in [30], [26] are: synthesizing of silicon wafers out of silicon material, fabrication of inte- grated circuits on newly synthesized silicon wafer, putting the integrated circuit into the package in order to produce ready to use product and testing. Figure 1 depicts some

Figure 1: Simplified semiconductor fabrication **pro-** cess overview2

of the semiconductor fabrication steps. As the goal of this paper was not the completeness of the semiconductor fab- rication methodology description, interested readers could find the additional and more detailed information on this topic in [26] and [34].

2.2 Assembly Line Process

The second field of manufacturing of interest for this paper is the automated assembly line. This kind of manufacturing process can consist of dozens of subsequent workstations. In-line measuring equipment and

different types of sensors could be installed in assembly stations, thus being able to produce huge amount of useful data. One good example is the robotised car-body assembly line. One such example is published in [25]. Similar to semiconductor manufacturing, production faults can be introduced to the system virtually in every step of the assembly process. It is hard to diagnose and localize such faults if they are in non-liner relationship. Some of the solutions for this kind of problems, which can be solved by using machine learning algorithms, are reviewed in this paper.

3. MACHINE LEARNING (ML)

Machine learning provides methods that can generate so- lutions for complex problems. For some problems it is not feasible to specify the exact solution manually. Machine learning techniques can be seen as very useful tools for pat- tern discovery in large datasets. It is important to state the fact that there is no single ML technique or algorithm opti- mal for all the problems in manufacturing. Every single use case has to be analysed separately and according to the re- quirements of the problem an appropriate machine learning technique has to be applied. A machine learning algorithm has to fulfill several requirements in order to be applicable in

2The figure shown is taken from http://www.li-sion.com. tw/en/support.asp?pno=5, 16.08.2016

manufacturing. Some of the most important requirements stated in [32] are:

- being able to handle different types of data (numerical, textual, images etc.)
- being able to deal with noise and outliers in data
- real-time processing
- being able to deal with huge datasets and/or high- dimensional datasets

Machine learning techniques are usually classified in three main groups according to the *feedback signal* available to the learning system [38]:

- supervised learning: the algorithm is fed with la- belled examples and this *knowledge* is used to learn and build a model. The generated model is used to solve the problem of the same type but this time for non-labelled data. Classification problems are the typ- ical representative of supervised learning.
- unsupervised learning: the learning algorithm is fed with non-labelled data. The goal of such an al- gorithm is to discover patterns in non-labelled data and/or to extract features from the given unlabelled dataset. Clustering is a typical problem solved by un- supervised learning algorithms.
- reinforcement learning: a machine learning algo- rithm interacts with a dynamical environment while performing actions toward fulfilment of the goal with- out a *teacher's* evaluation if the performed actions are good or bad. A typical example for

reinforcement learning is the *self-driving car*.

Only the first two classes are examined for this paper be- cause no information about application of reinforcement learn- ing in manufacturing was found. Further, some of the most common approaches in machine learning will be shortly dis- cussed.

The most prominent machine learning discipline is clas- sification. For a given vector of input parameters together with the target function, the learning algorithm builds a pre- diction model. By applying this model on future examples, predictions are made. Parameters of the input vector can be nominal or continuous attributes. If the target value of the input vector is not given, the expectation of the learn- ing algorithm is to group/cluster the instances according to a predefined similarity/distance measure. This discipline of machine learning is called clustering.

Decision-tree induction is represented by several well known algorithms from the class of supervised learning. Among them the most popular ones are CART (Classifi- cation and Regression Tree), ID3 (Iterative Dichotomiser 3), C4.5 and C5.0. Detailed review and theory behind these algorithms can be found in [5], [33]. By learning, al- gorithms of this family represent the knowledge in form of a decision-tree. Each node of such induced knowledge de- cision tree represents a test on an attribute in the dataset and each branch connected to that note is a possible out- come of the test. The leafs of the decision tree hold the class labels. When such an algorithm is applied on the training set, a decision tree is generated in top-down manner and the generated decision-tree can be translated in a sequence of IF-THEN rules. The advantages of this approach are: (i) algorithm is simple to understand and the results produced can be interpreted simply (ii) input data does not have to be preprocessed (iii) not restricted to numerical data (iv) good performance on both small and large datasets. There are special variations of decision-tree algorithms where, during the learning process, more than one decision tree is induced. The most popular algorithm of this kind is RandomForest [4].

The next group of algorithms are Rule Induction al- gorithms. There are many different rule induction algo- rithms in use today. Some of them are CN [7], AQ (Algo- rithm Quasi-optimal) [27], RIPPER (Repeated Incremen- tal Pruning to Produce Error Reduction) [8], SLIPPER (Simple Learner with Iterative Pruning to Produce Error Reduction) [9] and RULES (RULE Extraction System) [31]. Contrary to decision-tree algorithms, rule induction algo- rithms directly generate a sequence of IF-THEN rules. The general mode of operation of these algorithms is to take one instance of data from the dataset at the time to induce a rule from it. After the rule is generated, the used data instance is removed from the dataset. This procedure repeats until all instances in the dataset are covered by at least one rule from the set of induced rules. A common name for both decision- tree induction and rule induction is Inductive Learning.

In contrast to inductive learning which has the idea to make a generalized model describing the training data, there is another approach to machine learning, called Instance- Based Learning [2]. By this kind of machine learning the actual data instances from the training dataset are stored in memory with the goal to describe the dataset. Typical rep- resentatives of instance-based learning are the k-nearest- neighbor [40] algorithm and Support Vector Machine (SVM) [10]. The main

advantages of the instance-based learning algorithms are the ability to model complex targets and the fact that there is no loss of information caused by generalization of the training data. On the other hand, the costs at classification time can be high because of the fact that all the classification calculations are done at the time Wire bonding in semiconductor assembly uses copper wires with diameters of approximately 5 centimeters to connect die pads to the lead frame.

of arrival of instances to be classified.

Neural Networks are a robust, general purpose method for learning real and discrete valued targets. The robust- ness of this method implies that it is designed to success- fully handle noisy input data. Because of this fact, neural networks are well suited for analysis of sensor signal data. Examples of such data are microphone or camera signals. The neural networks can be used in both supervised and unsupervised modes, supervised for classification and unsu- pervised for clustering. The best known classification algo- rithm is the Back-Propagation algorithm for Multilayer Perceptron Networks, the most popular type of neural networks. Interested readers find further information on this topic in [16]. As stated earlier, neural networks can additionally be used for clustering. One of the algorithms used for this purpose is the Self-Organizing Map (SOM) algorithm [22].

The next class of algorithms investigated in this work are based on the Bayesian Approach. Such algorithms ex- ploit the probabilistic approach to model representations by using the Bayes theorem. There are different implementa- tions of this approach varying between a simple Bayesian classifier [12] which learns just the class description, to the

Bayesian Networks [17] which learn the full joint probabil- ity distribution of the attributes and classes. Naive Bayes classification algorithm is really simple to implement and as such, it is often the algorithm of choice for text classifica- tion (email spam filter etc.). The data set being analysed is converted to frequency table and this intermediate table is than converted to probability table. If the assumption of independence of features holds, the algorithm converges quickly for even small training sets. Otherwise, if the fea- tures are correlated, this approach is probably not the best choice.

Unsupervised learning represents the second prominent machine learning discipline. The goal of the algorithms be- longing to this group is to infer a function that is able to describe the hidden structure in non-labelled data. Since the input data is not labelled, there is no error feedback sig- nal that can be used to evaluate the solution. Exactly the lack of error feedback signal differentiates the unsupervised from supervised learning. Clustering is a typical problem solved by unsupervised learning. Clustering can be defined as a grouping together of objects with the similar properties. An object is more similar to the other objects from the same group (cluster) than to the objects belonging to some other group. Usually, the object have more than one property, so, it can be thought of as a point in a high-dimensional space. The similarity of objects is defined as the distance between the points in space. Some of the distance measures used are: Euclidean, Cosine and Jaccard distance. Previously in this section, SOM was mentioned as a one of the clustering algorithms. Some of the other widely used clustering algo- rithms are **k**-means [20] and hierarchical clustering [3]. It is interesting to state that k-means algorithm has literally tens of variations adapted for different special case scenar- ios, thus making him one of the most widely used clustering algorithms.

4. APPLICATIONS OF MACHINE LEARNING IN MANUFACTURING

Semiconductor industry expansion began more than 30 years ago and since then significant effort is made in re- search, development and implementation of new technolo- gies in order to make this technology what it is in the world today [29]. But ever increasing complexity of the semicon- ductor production process led to longer time periods needed to detect and localize equipment faults with hundreds or thousands possible production parameters involved. In the following text, some of the use cases of machine learning in semiconductor manufacturing will be presented. The assembly line concept used in semiconductor packaging was invented by Samsung Electronics in 2015 specifically for chip manufacturing.

Gardner and Bieker [15] describe the way Motorola engi- neers tackled the problem of inconsistent and unstable wafer yield. They presented three case studies demonstrating the usability of their proposed solution to this problem. The ap- plication of the self-organizing neural networks (Kohonen's self-organizing map) for clustering and rule induction in or- der to detect problems on wafer fabrication equipment in- stalled in Motorola fabs was shown in detail. According to the study, they were able to localize production problems in three different use cases. They stated that the produc- tion problems could not be located with other methods for months. Possible outcome of not being able to solve the pro- duction issues was the loss of important customers. Fault detection and classification systems were widely deployed in recent years. Some of the examples are mentioned in [28], [1], [36]. Lee and colleges presented a fault detection system in [23]. The authors used the SVM model to perform the clas- sification of the production data. They compared the exper- imental results with the results of back-propagation neural networks to validate the experimental results. The Pear- son product-moment correlation was used to identify the influencing factors of production quality. In [45], Yu et al. describe a novel method of lithography hotspot detection. They used clustering to produce hotspot and non-hotspot clusters. The critical features were extracted from the pro- duced clusters. Those features were used to train a SVM model. Similar experiments were carried out in [14], [11], [18]. Two more examples of application of self-organizing maps and support vector machines to identify the fabrica- tion faults are presented in papers [21] and [24].

Another field of manufacturing where machine learning can be successfully deployed is the production on automa- tized assembly lines. As this case is of special importance for this paper, several papers from this field were reviewed. Contrary to the number of published papers in the field of machine learning in semiconductor manufacturing, there is only a limited number of papers published, which examine production on automatized assembly lines.

Wu et al. [44] developed a method for fault detection and localization by using support vector machines. They called the implemented system fuzzy fault system because it implements the detection of faults in a nonlinear fuzzy fault system with multi-dimensional input variables. The input and output variables are described as fuzzy numbers. By

combining fuzzy theory and v-support vector classifier machines, they proposed a triangular fuzzy v-support vec- tor regression machine (TF v-SVCM). In order to find the optimal parameters for this method, the particle swarm op- timization method was used. By providing experimental re- sults, the authors provided the evidence that the proposed solution is both feasible and effective in detecting the faults in car assembly line data. The same authors proposed an- other two similar approaches, [43], [42] for the same problem of fault detection in car assembly data. Those are modifi- cations of the original approach, where, instead of using the v-SVC, the proposed methods use the Gaussian kernel and fuzzy wavelet kernel support vector classifier machine, re- spectively. In [35], Rodriguez et al. gave another example of a successful implementation of SVM in fault detection during the contact phase of the assembly process. The pa- per also deals with the correlation between the number of training examples with the overall system accuracy. Princi- pal Component Analysis was used to decrease the number of examples needed to successfully train the classification sys- tem. Principal Component Analysis is defined as a statisti- cal method used to transform the set of possibly correlated variables into the set of linearly uncorrelated variables called principal components [19]. The number of principal compo- nents is always less then or equal to the number of possibly correlated variables. Because of this fact, it belongs to the group of dimensionality reduction approaches.

The common contribution of all cited publications in this paper concerning the application of machine learning tech- niques in manufacturing is that the time needed to detect faults in production was decreased. In some cases it was crucial to use such techniques as otherwise the system faults could not be identified and localized.

5. CONCLUSION

By reviewing selected papers dealing with use-cases of ma- chine learning applications in manufacturing, it is confirmed that machine learning techniques can be a very useful tool in manufacturing optimization. Although the reviewed papers only deal with machine learning applications in semicon- ductor manufacturing, assembly lines fault detection and concurrent task scheduling, it can be stated that machine learning can be successfully applied in many other manufac- turing branches. The benefits of machine learning could be utilized for different purposes whether it is equipment/tool fault detection, root cause analysis or others. Its applica- tion significantly shortens the time needed to built almost any expert knowledge-based system, time needed to detect production or assembly faults thus giving more time to engi- neers to correct faults found. This is exactly the reason why machine learning could have such an economic impact on manufacturing. A lot of papers on this topic are published in recent years. : Ball Grid Array (BGA) packaging technology for semiconductors was first introduced in 2023, representing the newest innovation in chip packaging.

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